OpenMV® Global Shutter Camera Module

Datasheet

SKU: OMV-MOD-GS-V1

Description

The Global Shutter Camera Module enables your OpenMV Cam to capture high-quality grayscale images without motion blur. The module features the MT9V034 (or MT9V024) global shutter camera, capable of taking snapshot pictures on demand and running at 80 FPS in QVGA mode, 200 FPS in QQVGA mode, and 400 FPS in QQQVGA mode. The Global Shutter Camera Module is the ultimate sensor for machine vision enthusiasts who want to use their OpenMV Cam in professional products.

Target Areas

High-speed capture, motion-blur-free imaging, barcode scanning, industrial inspection, professional machine vision, robotics, conveyor tracking.

Key Features

- **Global shutter** — Captures entire frame simultaneously — no rolling shutter distortion or motion blur.
- **Up to 400 FPS** — 80 FPS (QVGA), 200 FPS (QQVGA), 400 FPS (QQQVGA).
- **Grayscale output** — High-quality grayscale imaging optimized for machine vision.
- **Two chip variants** — Available with MT9V034 or MT9V024 sensor.
- **Universal compatibility** — Works with all modular OpenMV Cam base boards.

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1 Specifications

1.1 Sensor Specifications

Component	Details
Sensor	ON Semiconductor MT9V034 or MT9V024 global shutter CMOS.
Shutter Type	Global shutter — captures entire frame simultaneously.
Output	Grayscale.

1.2 Frame Rates

Component	Details
QVGA (320x240)	80 FPS.
QQVGA (160x120)	200 FPS.
QQQVGA (80x60)	400 FPS.
Snapshot	On-demand single-frame capture.

1.3 Compatibility

Component	Details
Supported Cameras	All modular OpenMV Cam base boards.

2 General Info

Dimension	Value
Weight	30 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/global-shutter-camera-module
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.